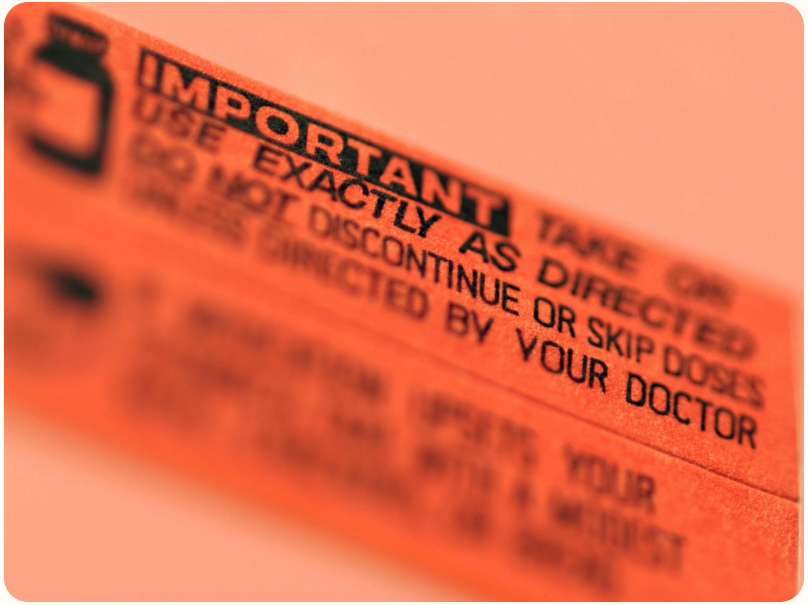
Medical Text Simplification Using NLP

Hana Muhsin, SAT 5114

Introduction



Medication instructions are frequently written at a **very high reading level**, often exceeding the average reading grade level in the U.S.

This complexity leads to **misunderstandings** of critical healthcare information.

GOAL: Simplify text while preserving original meaning.

Defining The Problem

High Complexity

Medical text is often written at a 12th grade level or higher, making it inaccessible for many patients.

Patient Risk

High-level language leads to misunderstood dosages and warnings, creating significant health risks.

Semantic Integrity

Simplification efforts must not distort the underlying medical meaning or clinical context.

Detail Preservation

Critical details like specific doses and safety warnings must be maintained during the simplification process.

Architecture & Pipeline

01 Collect medication text from FDA DailyMed (20 drugs)

02 Used GPT-4o to simplify text

03 Measured meaning using Sentence Transformer (cosine similarity)

04 Measured readability using Flesch-Kincaid scores

05 Measured Numerical loss for safety

Metrics Used For Simplification

Meaning Score(0-1)

- Uses a Sentence Transformer model and cosine similarity to measure how much meaning is preserved
- Goal: close to 1 (higher=better meaning preservation.

Reading Grade Level

- Flesch-Kincaid scale. Measures reading level required.
- Lower grade = easier reading.
- Target: ~6-8th grade

Reading Ease (0-100)

- Flesch-Kincaid Reading Ease.
- Higher score = easier to read.
- Low/negative scores = very complex text.

Numeric Loss

- Safety check to ensure critical numbers (dosages, counts) weren't lost.

Why These Metrics Matter

Grade Level

Shows if text became easier to understand for the target audience.

Reading Ease

Confirms quantitative improvement in overall readability.

Meaning Score

Sentence Transformer converts both original and simplified text into embeddings.

Then cosine similarity is used to compare them.

Captures overall meaning, not detailed clinical precison

Numeric Loss

Critical safety check to ensure dosage and patient safety.

How The Code Works (src):

1. Cleaning the text

- Normalize formatting
- Extract numeric tokens to preserve dosages and safety values

2. Simplifying (Strict Rules)

- GPT-4o with strict prompt that enforces
  - Lower reading level
  - Simple language
- Temperature = 0 for consistent output
- Main simplification function = simplify_text()
- Enforces readability + safety rules

3. Check Numbers & Final Result

- Compare before vs after; Retry for numeric preservation

Prompt Rules (Example)

- Keep all numbers exactly the same
- Do not change dosages or ranges
- Use simple words (6-8th grade level)
- No added medical information
- Strict enforcement of numeric preservation

Example of model output (Before vs After Simplification

ORIGINAL TEXT

Dosing for Adult and Pediatric Patients > 3 Months of Age

Treatment should be continued for a minimum of 48 to 72 hours beyond the time that the patient becomes asymptomatic or evidence of bacterial eradication has been obtained. It is recommended that there be at least 10 days' treatment for any infection caused by Streptococcus pyogenesto prevent the occurrence of acute rheumatic fever. In some infections, therapy may be required for several weeks. It may be necessary to continue clinical and/or bacteriological follow-up for several months after cessation of therapy.

SIMPLIFIED TEXT

How to take it:

- Take medicine for at least 48 to 72 hours.
- Continue until you feel better.
- Take for at least 10 days for Streptococcus pyogenes.
- Some infections need treatment for several weeks.
- Follow-up may be needed for months after stopping.

Key numbers (48-72 hours, 10 days) were preserved while improving readibility

Core Implementation Codes

Step 1: Text Simplification Logic

```
# Apply simplification to create a new column
df["Simplified_Text"] = df["Original_Text"].apply(simplify_text)
```

Step 2: Transformer-Based Meaning Measurement (Cosine Similarity)

```
# Load transformer model for semantic similarity
from sentence_transformers import SentenceTransformer, util
import torch
#load the model
model = SentenceTransformer('all-mpnet-base-v2')

#encode the original and simplified texts
original_embeddings = model.encode(df["Original_Text"].tolist(), convert_to_tensor=True)
simplified_embeddings = model.encode(df["Simplified_Text"].tolist(), convert_to_tensor=True)

# Compute cosine similarity as meaning score
df["Meaning_Score"] = [float(util.cos_sim(original_embeddings[i], simplified_embeddings[i]).item()) for i in range(len(df))]
```

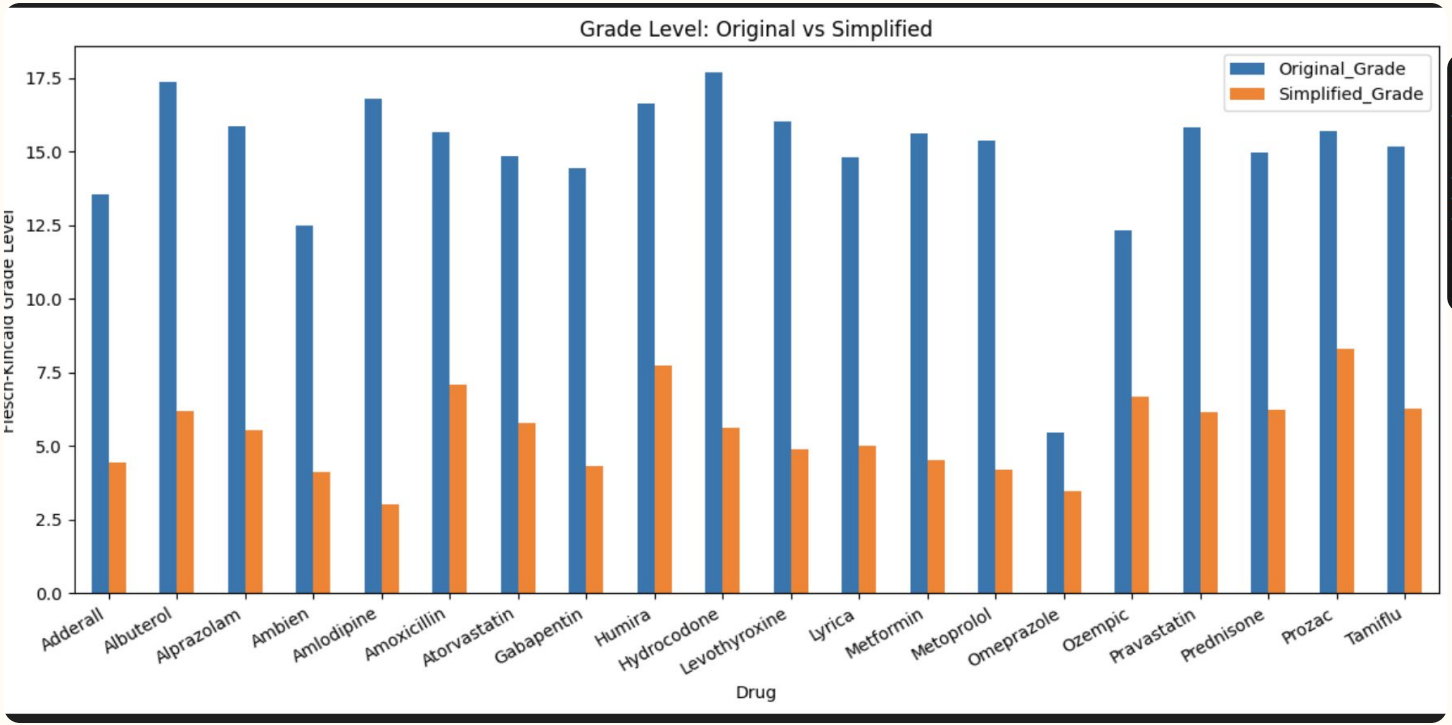
Code Implementation Continued

Step 3: Calculating Readability Metrics (Flesch-Kincaid)

```
import textstat
df["Original_Grade"] = df["Original_Text"].apply(textstat.flesch_kincaid_grade)
df["Simplified_Grade"] = df["Simplified_Text"].apply(textstat.flesch_kincaid_grade)

df["Original_Reading_Ease"] = df["Original_Text"].apply(textstat.flesch_reading_ease)
df["Simplified_Reading_Ease"] = df["Simplified_Text"].apply(textstat.flesch_reading_ease)
```

Grade Level Reduction




```
import matplotlib.pyplot as plt

ax = df.groupby("Drug")[["Original_Grade", "Simplified_Grade"]].mean().plot
kind="bar", figsize=(12, 6))

ax.set_title("Grade Level: Original vs Simplified")
ax.set_ylabel("Flesch-Kincaid Grade Level")
ax.set_xlabel("Drug")
plt.xticks(rotation=30, ha="right")
plt.tight_layout()

plt.show()
```

Key Findings

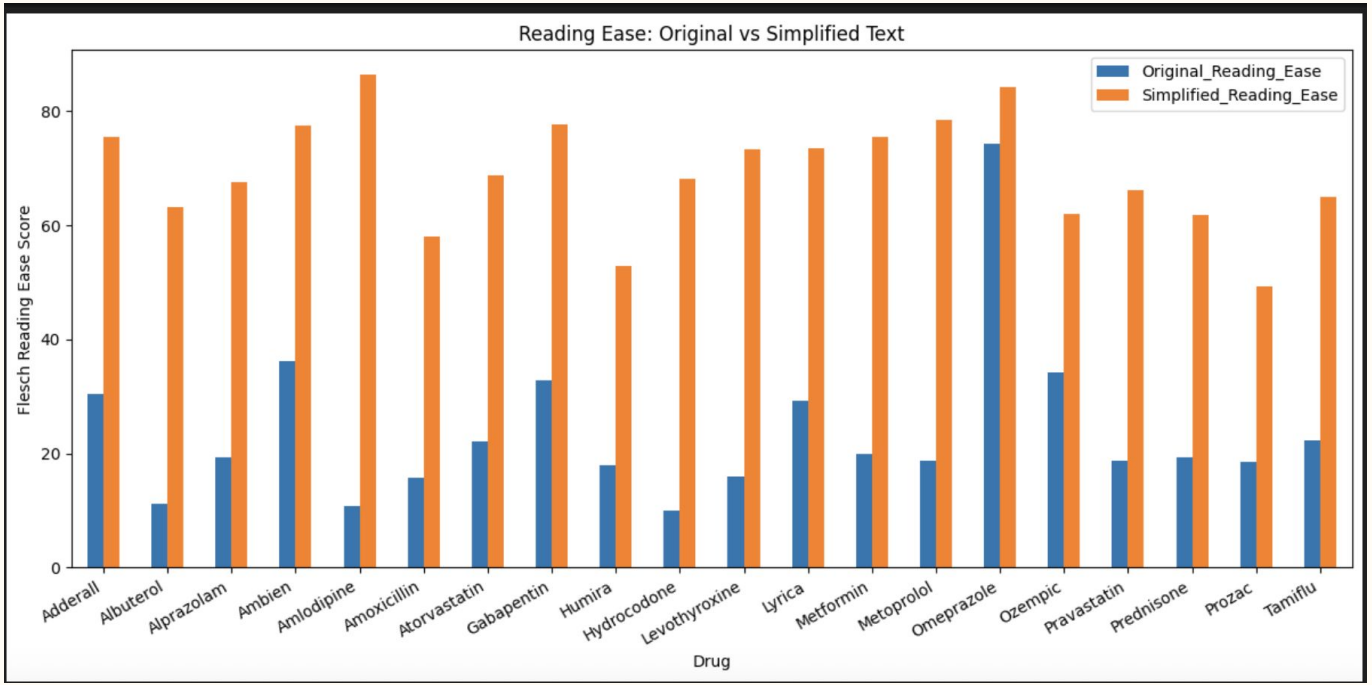
Significant Drop

Grade levels decreased across all medication samples.

Target Met

Simplified text reached approximately 4th-8th grade reading level.

Reading Ease Improvement



Key Findings

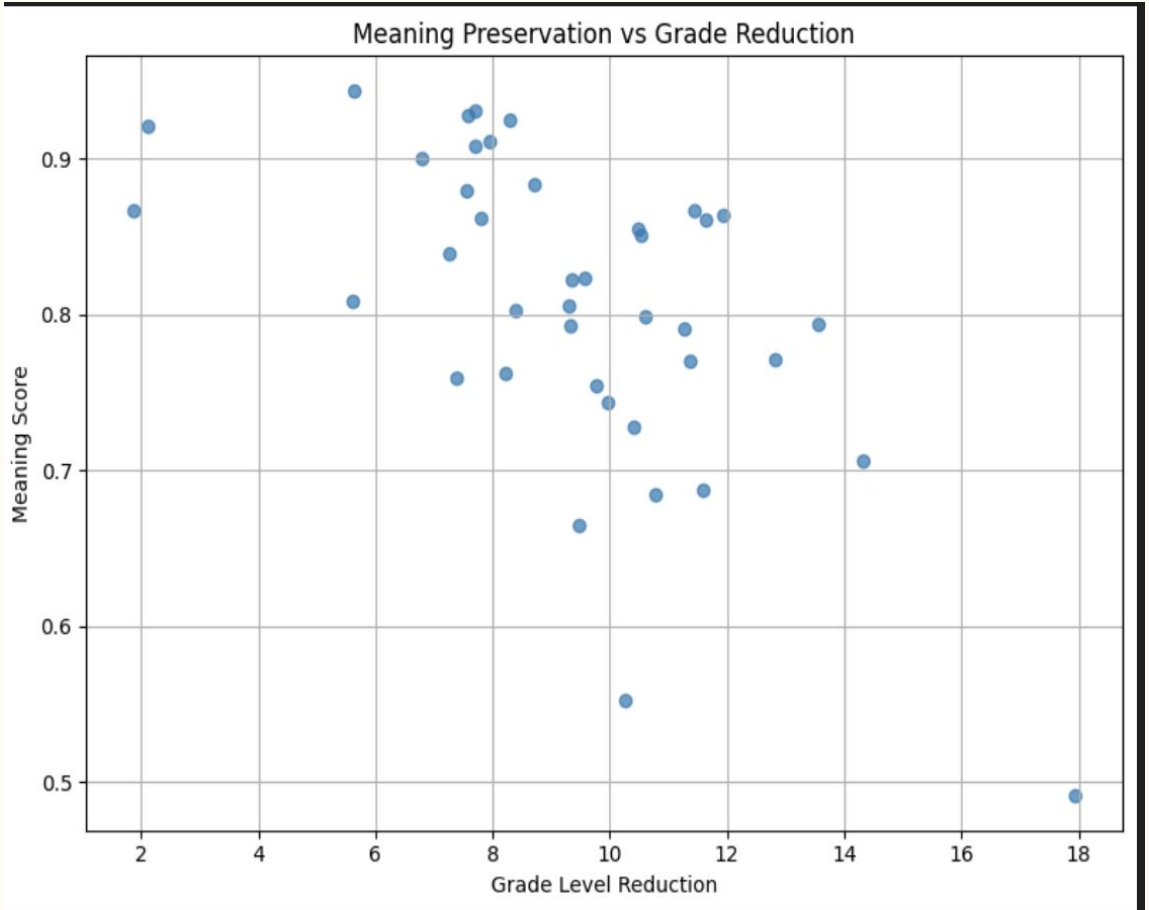
Increased Accessibility

Reading ease scores increased significantly across all medication samples.

Simplified Content

The simplified text version consistently scored higher, making it easier for patients to understand.

Meaning vs Grade Reduction




```
# Calculate grade reduction
df["Grade_Reduction"] =df["Original_Grade"] -
df["Simplified_Grade"]

# Plotting results
plt.scatter(df["Grade_Reduction"],df["Meaning_Score"],
alpha=0.7)
plt.grid(True)
```

Key Insights

Weak Correlation

Meaning preservation scores show high variance across different grade level reductions. Some reduction in meaning at higher levels of simplification

High Fidelity

Most samples maintained a meaning score above 0.7, even with substantial simplification.

Numeric Loss Analysis

Detection & Scoring

- Some numeric loss was detected, but most key values were preserved
- Numeric loss did not always correlate with meaning score

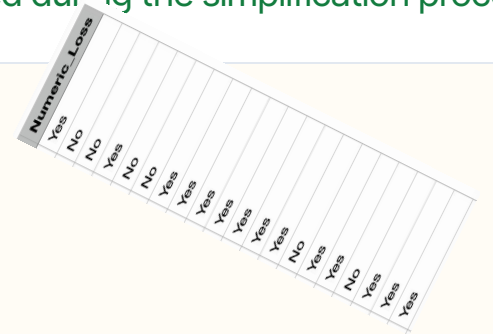
Manual Verification

Important values were mostly preserved upon manual inspection.

Primary Objective

Ensure dosages and timings were correctly maintained during the simplification process.

```
import re
def check_numeric_loss(row):
    original_numbers = set(re.findall(r'\d+', row["Original_Text"]))
    simplified_numbers = set(re.findall(r'\d+', row["Simplified_Text"]))
    return (original_numbers) != simplified_numbers
df["Numeric_Loss"] = df.apply(check_numeric_loss, axis=1)
```



Numeric_Loss
Yes
No
No
Yes
No
No
Yes
Yes
Yes
Yes
Yes
Yes
Yes
No
Yes
Yes
No
Yes
Yes
Yes

Summary of Results

final_results_table

Drug	Section	Original_Grade	Simplified_Grade	Original_Reading_Ease	Simplified_Reading_Ease	Meaning_Score	Numeric_Loss
Adderall	Dosage	11.4	3.7	44.6	82.2	0.9	Yes
Adderall	Warnings	15.8	5.3	16.2	68.6	0.9	No
Alprazolam	Dosage	13.9	4.5	27.0	74.9	0.8	No
Alprazolam	Warnings	17.9	6.6	11.5	60.1	0.8	Yes
Metoprolol	Dosage	12.9	3.4	35.0	84.4	0.7	No
Metoprolol	Warnings	17.9	5.0	2.4	72.6	0.8	No
Ozempic	Dosage	9.3	3.7	52.4	84.6	0.9	Yes
Ozempic	Warnings	15.3	9.7	16.1	39.4	0.8	Yes
Humira	Dosage	16.0	5.5	23.0	68.7	0.9	Yes
Humira	Warnings	17.3	9.9	12.9	37.1	0.8	Yes
Ambien	Dosage	9.5	2.7	50.1	86.1	0.9	Yes
Ambien	Warnings	15.5	5.5	22.5	68.9	0.7	Yes
Tamiflu	Dosage	14.2	6.7	31.8	61.8	0.9	Yes
Tamiflu	Warnings	16.1	5.9	12.7	68.2	0.6	No
Hydrocodone	Dosage	16.6	3.1	14.6	86.1	0.8	Yes
Hydrocodone	Warnings	18.8	8.2	5.3	50.0	0.8	Yes
Pravastatin	Dosage	15.9	4.3	23.4	82.0	0.9	No
Pravastatin	Warnings	15.8	8.1	14.2	50.1	0.9	Yes
Lyrica	Dosage	13.4	5.1	42.5	74.7	0.9	Yes
Lyrica	Warnings	16.2	4.9	16.0	72.1	0.8	Yes
Amoxicillin	Dosage	13.3	5.5	34.7	70.0	0.9	Yes
Amoxicillin	Warnings	18.0	8.7	-3.1	46.0	0.8	No
Metformin	Dosage	13.5	2.7	34.7	88.9	0.7	No
Metformin	Warnings	17.8	6.4	5.1	61.9	0.9	Yes
Omeprazole	Dosage	5.3	3.2	78.2	86.3	0.9	No
Omeprazole	Warnings	5.7	3.8	70.3	82.0	0.9	No
Albuterol	Dosage	14.9	4.5	30.6	76.3	0.7	Yes
Albuterol	Warnings	19.8	7.9	-8.1	50.1	0.9	No
Amlodipine	Dosage	12.4	2.8	33.0	86.4	0.8	No
Amlodipine	Warnings	21.2	3.2	-11.5	86.3	0.5	No
Prednisone	Dosage	13.7	4.4	29.0	75.6	0.8	No
Prednisone	Warnings	16.3	8.0	9.9	48.0	0.8	No
Levothyroxine	Dosage	13.5	5.5	33.2	69.2	0.9	Yes
Levothyroxine	Warnings	18.6	4.3	-1.3	77.5	0.7	Yes
Atorvastatin	Dosage	13.5	5.1	33.7	75.6	0.8	No
Atorvastatin	Warnings	16.2	6.4	10.7	62.0	0.8	No
Gabapentin	Dosage	13.0	4.2	43.3	78.9	0.9	No
Gabapentin	Warnings	16.0	4.4	22.5	76.6	0.7	Yes
Prozac	Dosage	14.7	7.5	23.1	56.3	0.8	Yes
Prozac	Warnings	16.7	9.1	14.1	42.5	0.9	Yes

```python
df_results = df[[
    "Drug",
    "Section",
    "Original_Text",
    "Simplified_Text",
    "Original_Grade",
    "Simplified_Grade",
    "Original_Reading_Ease",
    "Simplified_Reading_Ease",
    "Meaning_Score",
    "Numeric_Loss"
]].copy()

df_results["Numeric_Loss"] = df_results["Numeric_Loss"].map({True:"Yes", False: "No"})
```

Key Findings

Grade Level
Decreased across all medication samples.

Reading Ease
Scores increased, indicating improved accessibility.

Meaning Score
Remained consistently high (.70-.90).

CASE STUDY: Amlodipine Outlier Analysis

Original Text

Increased Angina and/or Myocardial Infarction Rarely, patients, particularly those with severe obstructive coronary artery disease, have developed documented increased frequency, duration and/or severity of angina or acute myocardial infarction on starting calcium channel blocker therapy or at the time of dosage increase. The mechanism of this effect has not been elucidated.

Simplified Text

How to take it:

- Some people may have more chest pain.
- This can happen when starting the medicine.
- It can also happen when increasing the dose.
- This is rare but can be serious.
- People with severe heart disease are at risk.
- The exact reason for this is not known.

Technical Complexity

Original has very long, complex sentence and technical medical language.

Structural Improvement

Simplified, short and clear statements. Simpler wording. Structured format.

CASE STUDY: Continued

Why this happens?

- Readability metrics (Flesch-Kincaid) are highly sensitive to sentence length and word complexity
- GPT simplifies based on prompt but doesn't strictly control the final reading level
- Small structural changes can impact readability scores.

Why the grade dropped

- Long sentence → multiple short sentences
- Complex terms → simpler words
- Structured format → easier readability

Meaning Preservation:

- Core meaning is preserved
- Some clinical detail is simplified
- Meaning score reflects overall similarity, not detailed clinical precision

Conclusion:

The results are influenced by the behavior of the model as well as the sensitivity of the readability metrics, and the structure and complexity of the input text.

Literature Comparison

Readability Focus (Picton)

NLP models are useful for improving readability, a finding well-supported in current literature. My results show the same pattern.

Simplification Tradeoffs (Paulson)

Tradeoffs between aggressive simplification and meaning preservation remain a key challenge. I observed a similar pattern, where higher grade reduction sometimes led to lower meaning scores

Results Consistency

My findings are consistent with prior studies

Data Scale Impact (Rahulprasath)

Another study showed that larger datasets improve performance

Small datasets limit generalizability.

Limitations & Future Work

Current Limitations

- Dataset is small (20 medications)
- Results may not fully generalize to all medical texts.
- Medical text varies in structure and complexity
- Readability metrics are highly sensitive to sentence structure and may inflate improvements
- Meaning score captures overall similarity but may miss small losses in detail.
- Different drug types may need different simplification approaches

Future Work

- Expand dataset and test on more diverse medical instructions.
- Validate the approach across different types of clinical text
- Develop more precise methods to evaluate clinical detail preservation
- Standardize preprocessing to ensure more consistent text structure
- Improve consistency across various types of text structures.
- Improve numeric preservation

Conclusion

Readability Improvement

Readability improved significantly across the simplified dataset.

Simplification effectiveness depends on structure and complexity of original text.

Meaning Preservation

Core meaning was mostly preserved despite grade level reduction.

Measured Tradeoffs

Extreme simplification introduced minor tradeoffs.

Results are influenced by both model and sensitivity of readability metrics.

Future Viability

Model showed strong results and high potential for future applications. This approach could improve patient understanding and help reduce medication errors.

RESOURCES

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Thank You

Hana Muhsin
hmuhsin@mtu.edu